

Youhan Huang

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🌐 [JohanHuang2005](#) 🌐 [Personal Website](#)

Research Interests: Computer Vision, Intelligent Sensing



Education

Beijing University of Posts and Telecommunications

Sep 2023 – Jul 2027

B.Eng. in Electronic Information Engineering

GPA: 92.05/100

Rank: 2/101

Core Courses: Engineering Mathematics (97), Linear Algebra (94), Advanced Mathematics (93), Computer Networks (97), Fundamentals of Programming (96), Introduction to Artificial Intelligence (95), Machine Learning (95)

Honors & Awards

National Scholarship (2023–2025), University Merit Student (2023–2025), Honorable Mention, Mathematical Contest in Modeling (MCM) (2025)

Research Experience

Project Lead | Multi-Task Histological Scoring for Mouse NAFLD (BUPT x Capital Medical University)

To address the demand for joint assessment of three tasks (steatosis, inflammation, and ballooning degeneration) in mouse NAFLD histological scoring, I led the construction and open-sourcing of a multi-task histological scoring dataset; proposed a joint modeling scheme based on Multi-task ViT, introducing task-specific Adapter/LoRA to mitigate negative transfer between tasks, and completed experimental design, model implementation and result analysis; participated in paper writing. **Co-first author (ranked 1st)** paper accepted as **Oral** at **ICME 2026 (CCF-B)** ([arXiv: 2605.29852](https://arxiv.org/abs/2605.29852)).

Research Intern | Intelligent Wireless Sensing and Application Research Center, Peking University

Aiming at the high-precision extrinsic calibration problem in cooperative multi-mmWave radar sensing, targeting the center estimation bias and missing pitch angle caused by existing methods treating the human body as a point target, I participated in constructing a consistent trajectory calibration scheme based on elliptic cylinder human modeling and geometric completion, achieving relative pose recovery between radars and pitch angle self-calibration; participated in key experimental design and result analysis. The method significantly reduces relative distance and azimuth errors, achieving centimeter-level distance error and stable angle estimation; the related paper has been submitted to **IEEE TMC (CCF-A)**.

For the low-bandwidth mmWave radar human skeleton sensing task, supporting the **SenSys 2027 (CCF-B)** research project. Responsible for building an end-to-end real-time sensing system, connecting the full visualization pipeline from raw radar signal acquisition, point cloud super-resolution processing to skeleton inference; completed reproduction and unified evaluation of three representative papers including m3Track and M4esh, establishing a SOTA comparison baseline; quantitatively verified the effectiveness of the SSE strategy in low-bandwidth scenarios by comparing point cloud quality and joint coverage across multiple bandwidth levels.

Research Intern | Pattern Recognition and Intelligent Vision Lab, School of AI, BUPT

To address the shape-appearance ambiguity caused by the coupling of reflection and refraction in 3D reconstruction

of transparent objects, I participated in research on polarization-based multi-view inverse rendering; participated in building a transparent object reconstruction framework, representing object shape with implicit geometry, introducing polarization illumination modeling and dual-interface light transmission modeling to jointly estimate object geometry and imaging process; participated in experimental implementation, result analysis and paper writing. The method outperforms existing state-of-the-art methods on both synthetic and real datasets; **first-author paper** has been submitted to **AAAI 2027 (CCF-A)**.

Research Intern | Image Aesthetics Assessment Lab, School of AI, BUPT

To address temporal distortion and standard drift of holistic scoring in long-video aesthetics assessment, I participated in the spatiotemporal quantitative assessment research project; responsible for SOTA baseline method reproduction, full pipeline construction of comparative experiments and multi-dimensional performance evaluation, participated in writing the experimental section and related work section of the paper; the related paper has been submitted to **IEEE TIP (CCF-A)**.

Project Experience

LidarCar: Autonomous Mobile Robot for Maze Navigation & Mapping (C)  [Repository](#) Jul 2025 – Nov 2025

Served as team leader, completed hardware system integration of an autonomous mobile robot equipped with 2D LiDAR for maze navigation and real-time mapping scenarios; responsible for LiDAR module integration, STM32 pin allocation and debugging, Bluetooth communication code development, and coordinated joint debugging of sensor, control and communication links, realizing core functions including maze exploration, exit localization and return-to-start.

International School TA Recruitment System (Java)  [Repository](#) Mar 2026 – May 2026

Served as team leader, designed and developed a TA recruitment system for course organizers, lecturers and graduate teaching assistants; responsible for requirement analysis, iteration planning, system architecture and core function implementation, advancing core workflows including job posting, application management, status tracking and result notification via Agile methodology and GitHub collaboration.

Multimedia Fundamentals Learning Website (HTML/CSS/JS)  [Repository](#) Mar 2026 – Apr 2026

Served as team leader, designed an interactive website for multimedia fundamentals learning; responsible for homepage information architecture, visual presentation and front-end implementation, and oversaw page maintenance, navigation consistency, cross-device compatibility and basic interaction optimization.

Skills

Programming: Python, Java, C, MATLAB, HTML
English: IELTS 7.5

Frameworks & Tools: PyTorch, Git, LaTeX, Linux

Interests

Football, Tennis, Travel, Reading